



ANMS 2026 · FIFTH INTERNATIONAL WORKSHOP ON AUTONOMOUS NETWORK MANAGEMENT SYSTEMS

DT-II

Ontologies, Multiagent Digital Twin and Intelligent Telemetry
for Internet Infrastructure

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UFABC · UFRGS · IFSPe · CGI.br — CGI-IETF Initiative

An Internet Too Big to Manage by Hand



79,441

active Autonomous Systems on the Internet

(CIDR Report, Geoff Huston / APNIC — September 16, 2026)

- Each AS is operated independently — different teams, tools, and vendor conventions.
- A single misconfiguration (e.g., a bad BGP announcement) can cascade across the global Internet.
- Day-to-day management still relies on manual procedures, ad-hoc scripts, and vendor-specific CLIs.
- Telemetry helps detect anomalies — but detection alone does not explain or govern them.

Telemetry Tells You “What” — Not “Why”



- Modern networks stream enormous telemetry — gNMI, SNMP, YANG-Push — across 5G, satellite, optical and BGP-level domains.
- But raw metrics carry no shared meaning across those domains: a “channel” in optical transport and a “channel” in 5G are not the same thing to a script.
- Detecting a symptom (packet loss, a flapping session) is not the same as understanding its cause, its blast radius, or its governance implications.

DT-II's premise: pair live telemetry with an explicit, machine-reasonable model of what things mean.

Ontology + Knowledge Graph



Ontology — the rulebook

Defines what a Node, a Connection or a Channel IS, and what must always hold true. The “DNA.”



Knowledge Graph — the live map

The actual routers, links and sessions running right now, as connected facts. The “organism.”

In DT-II: the ontology defines the logical framework (classes, relations, axioms); the Knowledge Graph is its continuously updated instantiation.

Three Ideas from Formal Ontology (UFO)



Kind — a fixed identity

A physical router is a router no matter how it's configured — identity that doesn't change.



Role — a context-dependent function

That same router is an “Edge Node” only while it plays that part in the topology.



Relator — a relationship treated as a thing

A BGP session or an SLA contract: it connects two entities and ceases to exist the moment either one does.

One Vocabulary for Every Domain



- The same three classes unify 5G, satellite, optical and BGP-level infrastructure — one “Logical Constitution” instead of per-technology models.
- OntoUML/OLED separates rigid identity-bearing types (Kind) from context-dependent functions (Role), enabling systematic validation and export to OWL/Turtle.

TBox, RBox, ABox



TBox — the rules

Class definitions & constraints. E.g., every NetworkNode must possess a logical identifier.



RBox — the relationship rules

Transitivity, property chains, mereology — how relations propagate and compose.

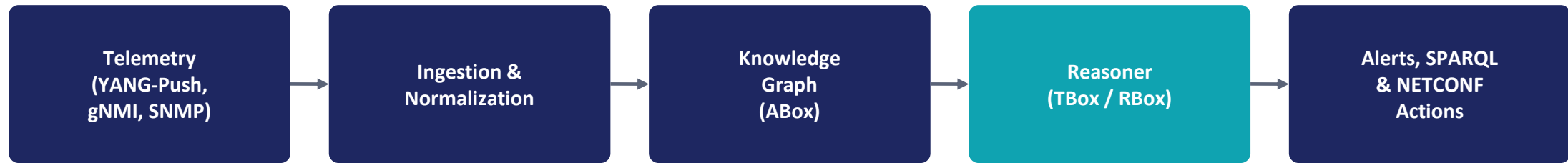


ABox — the live facts

Actual instances streamed in continuously from telemetry — the current state of the world.

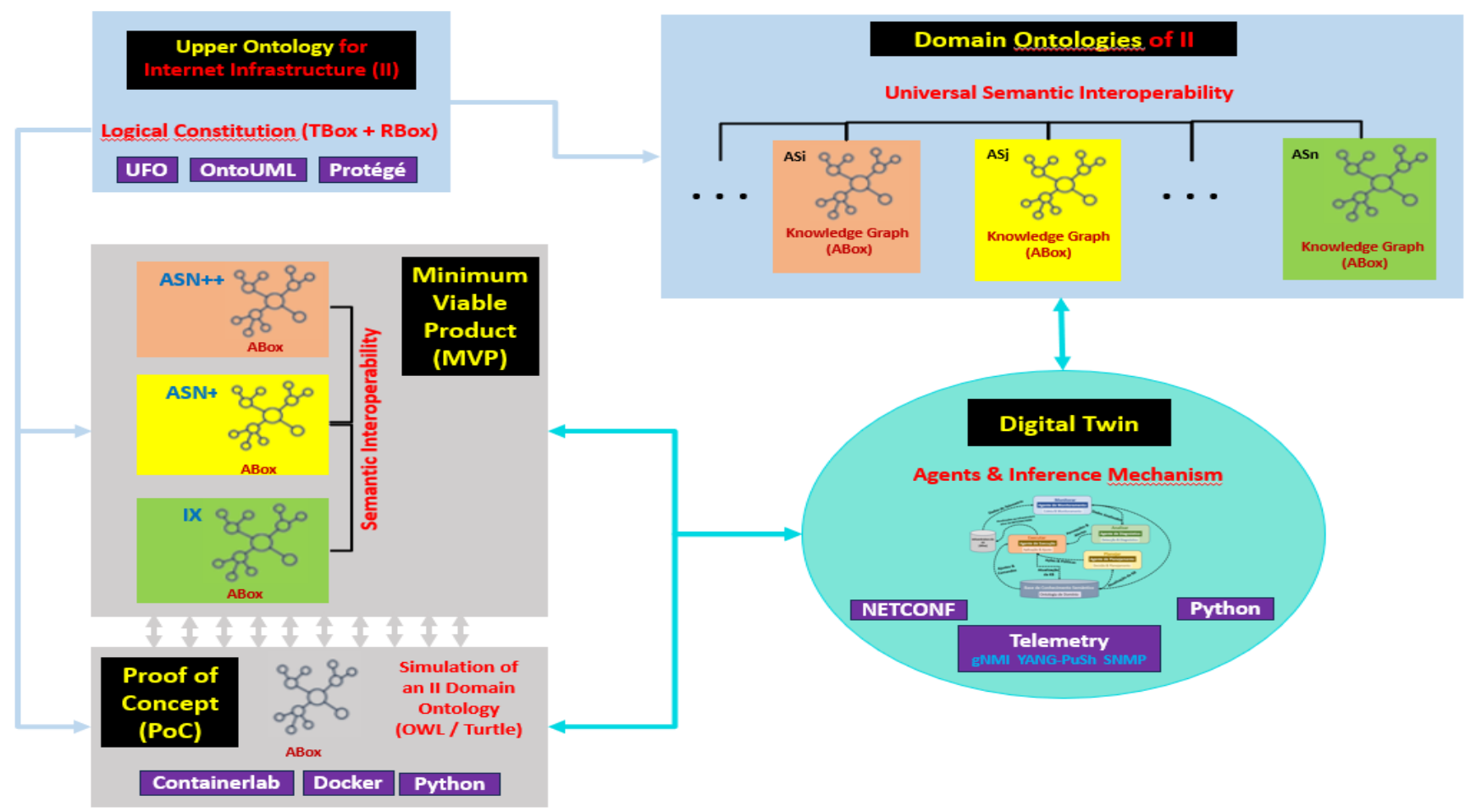
A reasoner checks all three together — e.g., an inactive Channel automatically implies a compromised Connection.

From Telemetry to Action

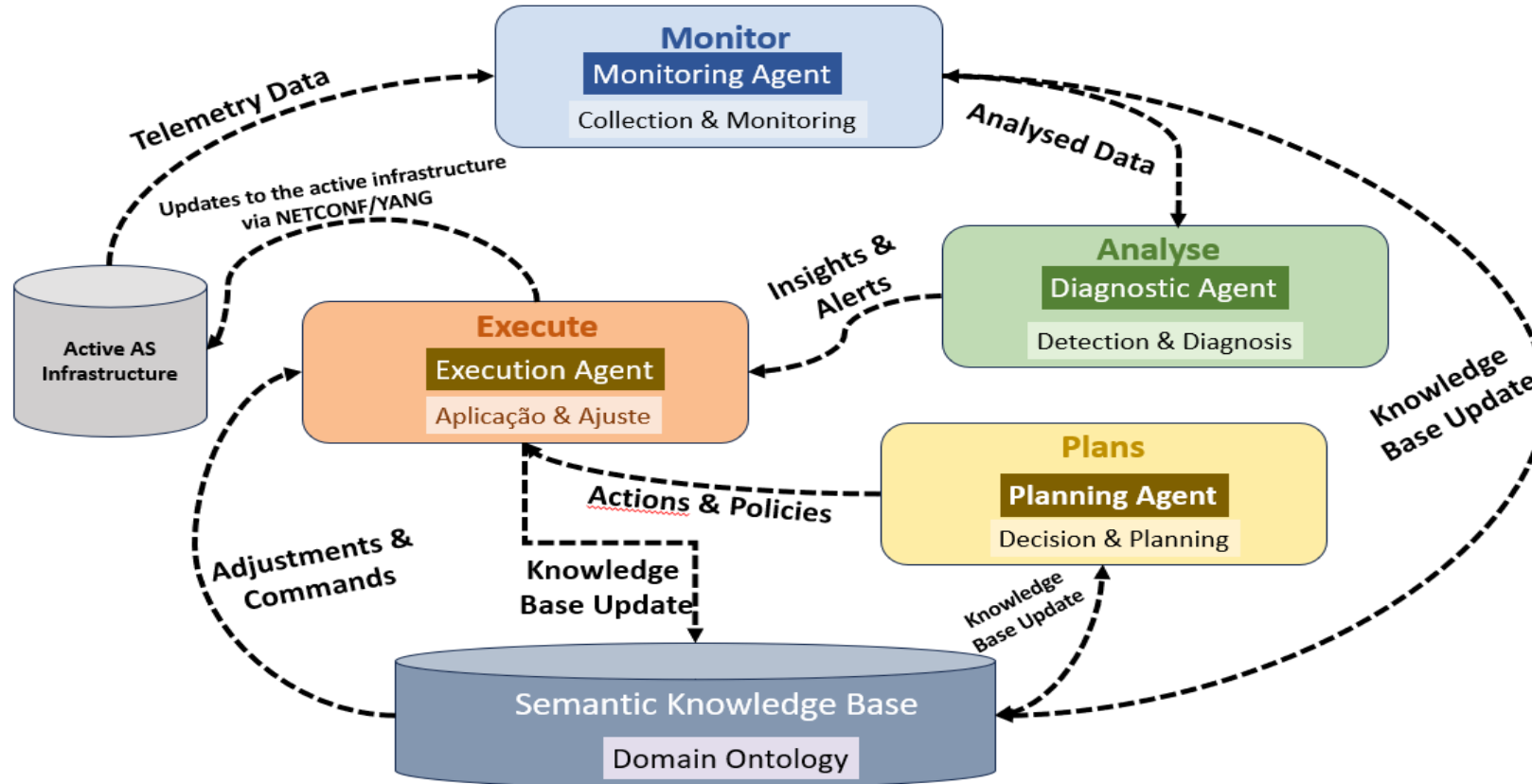


- Seven logical layers in the paper collapse to this pipeline: ontological, identity, ingestion, persistence, validation, reasoning, exposure.
- The same pipeline handles diagnosis (Sec. V) and governance (Semantic Jurisdiction) — only the axioms consulted differ.

Proof of Concept → Minimum Viable Product



MAPE-K: How the Twin Stays Autonomic



Ontology Meets Code



Example: a BGP session change updates the ABox — and the reasoner automatically derives `InactiveConnection`, with no rule written for that specific case.

- This loop is how the ontology actively guides code execution, not just documents it after the fact.



Governance Written Into the Ontology



- Jurisdiction is not a fixed hardware attribute — it is a Relator-mediated normative layer, defined by ontological axioms.
- Property chains in the RBox propagate legal and contractual constraints (LGPD, GDPR, SLAs) automatically across nodes, connections and channels.
- The result: compliance that a reasoner checks continuously — not a policy an administrator assumes is still true.

Enforcing Jurisdiction Automatically

1

Telemetry (gNMI, YANG-Push, SNMP) continuously populates the ABox with live topology and data flows.

2

The Reasoner asks: does this data flow cross a node whose jurisdiction fails to meet the required privacy constraint?

3

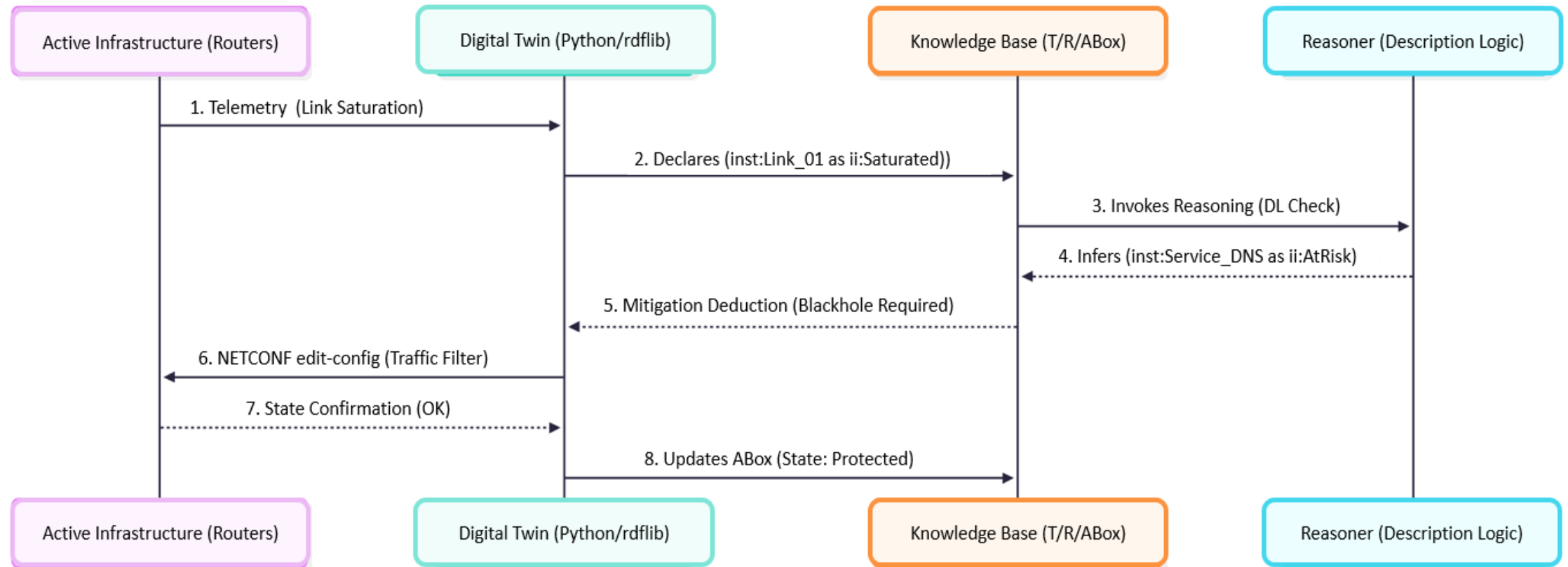
If violated, MAPE-K reacts: Analyse detects the violation, Plan searches for a compliant alternative.

4

Execute applies the change via NETCONF — only after logical validation against the Logical Constitution.



Mitigating a DDoS Attack in Milliseconds



Unlike blind IP-blocking: every mitigation step is explainable and validated against the ontology before it runs.

How DT-II Positions Itself

Typical Approaches

- Network Digital Twins driven mostly by metrics or learned policies.
- Network ontologies rich in concepts, rarely wired to live telemetry.
- ITU-T / IETF frameworks establish accepted high-level specifications, not executable reasoning.

DT-II

- An ontology-governed twin, explicitly validated against semantics — not only against metrics.
- Live YANG-Push telemetry feeding automated DL reasoning over TBox/RBox/ABox.
- Governance (Semantic Jurisdiction) as first-class, verifiable logic — not a procedural checklist.

Contributions at a Glance



A reference ontology for Internet Infrastructure: Node – Connection – Channel.



DL/OWL operationalization enabling automated reasoning over TBox, RBox and ABox.



A synchronization approach aligning the Knowledge Graph with model-driven telemetry (YANG-Push).



Semantic Jurisdiction: a governance mechanism for verifiable, explainable compliance.



A multi-agent Digital Twin architecture following the MAPE-K cycle for policy-driven action.

Where We Are Today



- This is a conceptual, design-stage contribution — not yet a large-scale deployment.
- Current reasoning uses full SROIQ expressivity, which carries real computational cost at Internet scale.
- Ontology evaluation via competency questions, and systematic performance benchmarking, are in progress rather than published results.

What's Next

1

Simulate the Digital Twin and evaluate alarms across Autonomous Systems.

2

Validate spatio-temporal response mechanisms experimentally.

3

Benchmark SROIQ reasoning against lightweight profiles (OWL 2 RL/RDFOx, OWL 2 QL/OBDA).

4

Document the ontology's competency questions and design rationale.

5

Extend semantic interoperability to 5G, satellite and SD-WAN.

6

Submit as an IRTF NMRG Internet-Draft within the CGI-IETF initiative (2026).



From Active Infrastructure to Autonomic, Legally-Aware Infrastructure

“Infrastructure that understands what it moves, why it moves it,
and whether it is allowed to.”

github.com/LJuliaoBraga/dtii

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Thank you — Questions?

